Data test

Hypothesis Testing Report

23 hypotheses tested · 19 rejected, 4 accepted-on-the-metric-that-logged-them, 0 survived independent confirmation

A deterministic, point-in-time rebuild of India-equity systematic signal testing – NSE bhavcopy 2004–2026, real T+1 execution and costs, 30-seed placebo floor, train/val discipline, and a 10,000-path Monte Carlo bootstrap.

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Executive summary

Between 2026-08-14 and 2026-08-19, this project tested 23 logged hypotheses built on 10 distinct economic ideas (mean reversion, delivery-confirmed breakouts, OI-confirmed breakouts, FII-flow-gated dips, volatility-contraction breakouts, wide-range price-action bars, correlation-screened and same-sector pairs trading, and 12-1 month cross-sectional momentum), each tested on one or both of two data splits and, where a result looked promising on train, re-tested on a held-out val window.

19 of 23 were rejected outright – the real mean return was either negative, or positive but ranked below the placebo band (blind selection from the same eligible pool on the same dates). **4 were mechanically "accepted"** by this project's own logging rule (beats every placebo seed, and positive portfolio Sharpe where a portfolio simulation exists) – **none of the four have survived independent scrutiny**: both same_sector_pairing variants failed their own val confirmation and then flipped sign entirely on the delivery split; momentum's delivery/train acceptance is undercut by its own delivery/val result, which rests on only 9 weekly buckets, a t-stat of 0.98, and a mean (+6.55%) driven by a handful of outliers against a negative median (-3.21%).

Plain NIFTY50 buy-and-hold (CAGR ~12.8%, Sharpe ~0.62–0.66) beat every mechanical rule tested on every window. A 10,000-path block-bootstrap Monte Carlo, added independently of the t-stat/placebo pipeline, corroborates this: the clean rejections stay at ~0% probability of a positive mean under resampling, and the same_sector_pairing collapse is visible from a second angle (99.9% on train → ~74% on val, confidence interval straddling zero).

The defensible conclusion, stated in this project's own working notes and unchanged by this report: short-horizon technical reaction to a visible price/delivery/OI/flow/volatility dislocation does not survive real execution and costs on this universe, at retail cost structure – not that nothing works, full stop. Longer-horizon momentum, fundamentals/valuation, and a genuine volatility/credit stress-regime gate remain under-explored, and are named explicitly in the Open Threads section.

Methodology

Five rules, each enforced by code and tests, not convention alone:

Deterministic	Seeds, content hashes, and run manifests. An AST scan fails the build if any module reads the wall clock outside two allow-listed metadata timestamps.
Point-in-time	No datum reaches a decision before it existed. The universe is a recomputable rule (monthly-rebalanced, banded top-200-by-turnover), never today's index membership projected backward.
Executable	Every fill is at the next session's open (T+1), never same-bar - a signal detected at bar T's close can only act on bar T+1.
Benchmark-relative	Every headline number is compared against NIFTY50 buy-and-hold, after real costs, on the same capital and dates.
Counted	Every hypothesis tried is logged to hypothesis_log.csv, append-only, whether it wins or loses - so a hit rate can never be computed over a silently-curated subset.

Trade simulation and evidence bar

Trade-level results are sizing-independent (fixed capital per trade, reported as percentages) - the right tool for asking whether a signal carries information, separate from portfolio-level results (Rs 50,000 capital, 5 concurrent slots, real drawdown), which answers whether it survives as an account. Confirmation-style signals (mean_reversion, delivery_breakout, oi_momentum, participant_tilt, vol_squeeze_breakout, price_action) use a 7-day max hold / 2.0x-ATR stop / 1:2.5 risk-reward exit rule, shared across all of them so results stay comparable; momentum uses a pure ~1-month calendar hold instead, matching its own construction. Every result is checked against 30 placebo seeds (same signal dates and counts, names drawn blindly from that date's eligible pool) - the noise floor a real result must clear - and significance is read off a non-overlapping entry-week bucket t-stat, since trades entered the same week substantially share one market draw and would otherwise overstate how much independent evidence exists.

A hypothesis is logged “accepted” only if it beats every placebo seed (and, where a portfolio simulation exists, has positive portfolio Sharpe) - this is a mechanical gate applied at logging time, not a claim of statistical confidence on its own; every acceptance in this project has still needed a val-window confirmation before being trusted, per the train-decides-first discipline below.

Train / validation discipline

A result is never promoted on the window it was discovered on. Val is touched only after a candidate clears train; test stays untouched throughout this whole program. Two splits are used, each with an embargo gap between windows:

Split	Train	Val	Test (untouched)	Notes
primary	2004–2016	2017–2021	2022–2026	Full price history, 60-day embargo
delivery	2019-06-27–2023-06-30	2023-07-01–2025-03-31	2025-04-01–2026-08-13	Per-stock delivery/OI/FII data starts 2019 – short history, higher evidence bar

Monte Carlo addition (2026-08-19)

Extends the same evidence pipeline with a block-bootstrap resample of the real, already-simulated trades - not a new backtest. For each hypothesis, the real trades are grouped into the same entry-week buckets the t-stat is built on, then resampled in contiguous 4-bucket (~1 month) blocks, circularly, 10,000 times, to build the sampling distribution of the mean under resampling. Two numbers are reported per hypothesis: the share of resampled histories with a positive mean (a bootstrap analogue of the t-stat), and an illustrative full-capital sequential-compounding read (no position limits - a magnitude sense only, not a portfolio claim).

Data foundation

Source	What	Coverage
NSE bhavcopy archive	Rebuilt fresh per-year parquet store (not reused from any prior project)	5,588 trading days, 2004-01-01–2026-08-13
fno.db	Per-stock futures/OI, read-only, content-hashed every run	48+ GB; OI/futures data from 2008
Corporate actions	Detected and adjusted directly from the price series	Full history
Delivery / participant-flow data	Per-stock delivery %, FII net index-futures positioning	From 2019-06-27 (the delivery split's own start)
NIFTY50 index level	Benchmark yardstick and this report's own bull/bear/sideways regime read	Full history
Universe	Point-in-time, monthly-rebalanced, banded top-200-by-turnover – never today's index membership	Recomputed every run

Signals tested

Ten distinct economic ideas, each with its own falsifiable story stated before any code was written:

mean_reversion

A stock pushed 1.5+ std devs below its own 50-day average in a short window is disproportionately a forced/panicked seller (margin calls, index rebalancing, tax-loss selling) rather than new information about impaired value - expected to partially correct once the selling pressure exhausts. The only predecessor-project strategy that survived a genuine walk-forward test; re-tested here under honest execution.

delivery_breakout

A price breakout on ordinary/low delivery is disproportionately intraday speculation that unwinds overnight. A breakout with delivery meaningfully above its own recent normal is disproportionately real buyers converting the move into settled positions.

oi_momentum

A breakout on falling/average open interest is disproportionately short-covering with no fresh leveraged conviction. A breakout with open interest rising meaningfully faster than normal means participants are opening new leveraged positions into the move.

participant_tilt

A mean-reversion dip bought while FII net index-futures positioning sits above its own recent trend is a normal pullback inside continued institutional accumulation; the identical dip bought while FII positioning trends down is the early stage of a real breakdown. Market-wide gate only (no per-stock FII breakdown exists) - it decides WHETHER a dip is bought that day, not WHICH stock.

vol_squeeze_breakout

A breakout following a genuine contraction (short-term range compressed well below its own longer-run normal) is the first real re-pricing after a quiet period, not noise inside an already-active range. Tests the volatility axis none of the price/delivery/OI/flow signals touched. A delay=2 variant was also tested, entering two sessions after the signal fires, to isolate whether buying at the peak of the dislocation (not the premise itself) explained the rejection.

price_action (LONG)

A session that is BOTH unusually wide-range AND closes pinned at one extreme, on volume well above normal, is disproportionately genuine new information or real institutional participation - unlike vol_squeeze_breakout, needs no prior contraction, only that today's bar is the outlier.

pairs_reversion (correlation-screened)

Two same-sector, historically correlated stocks whose log-price spread drifts unusually wide are disproportionately showing a temporary, idiosyncratic dislocation rather than a genuine re-rating - market-neutral by construction, needing only the relative mispricing to close.

same_sector_pairing

A looser version of the pairs premise: shared sector membership alone (without requiring the pair to already be historically correlated) is claimed to be enough linkage for a wide relative-price dislocation to be temporary. Tested with both a random-draw and a liquidity-ranked pair-selection rule, against a common any-sector-random placebo.

momentum (12-1 month)

A stock that outperformed over the trailing ~12 months (skipping the most recent month, to exclude the short-term reversal window this project's own entry-timing diagnostic found real) is more likely still mid-diffusion of genuine improving information than already fully priced. Long-only, top quintile of the point-in-time universe, re-ranked monthly.

Full test log – all 23 hypotheses

Every row is a real, logged run in hypothesis_log.csv. “n” is resolved trades, “buckets” is the independent entry-week count the t-stat is computed on, “Δ placebo” is real mean minus the mean of 30 placebo seeds.

#	Signal	Split	Window	n	buckets	mean %	t-stat	Δ placebo	Decision
H1	mean_reversion re-test	primary	train	20,499	616	-0.97%	-0.82	-0.36%	rejected
H2	delivery_breakout	delivery	train	2,194	170	0.08%	-1.95	-0.29%	rejected
H3	oi_momentum	primary	train	2,448	328	-0.63%	-4.52	-0.95%	rejected
H4	participant_tilt	delivery	train	2,916	125	-0.52%	2.00	-0.31%	rejected
H5	vol_squeeze_breakout	primary	train	4,792	558	-0.38%	-5.98	-0.63%	rejected
H6	vol_squeeze_breakout (delay=2, honest execution)	primary	train	4,802	550	0.08%	-1.77	-0.02%	rejected
H7	pairs_reversion	primary	train	197	149	0.15%	0.18	-0.10%	rejected
H8	price_action LONG	primary	train	8,807	618	-0.59%	-6.17	-0.64%	rejected
H9	pairs_reversion	primary	train	216	154	0.18%	0.58	-0.22%	rejected
H10	same_sector_pairing – random	primary	train	3,409	470	0.27%	3.49	0.18%	accepted
H11	same_sector_pairing – liquidity	primary	train	3,705	469	0.24%	3.70	0.09%	accepted
H12	same_sector_pairing – random	primary	val	2,068	249	0.05%	0.77	0.47%	rejected
H13	same_sector_pairing – liquidity	primary	val	2,301	249	0.07%	0.95	0.35%	rejected
H14	same_sector_pairing – random	delivery	train	1,739	210	-0.07%	0.37	0.17%	rejected
H15	same_sector_pairing – liquidity	delivery	train	2,021	209	-0.06%	0.53	0.14%	rejected
H16	vol_squeeze_breakout	delivery	train	1,725	194	-0.18%	-0.82	-0.33%	rejected
H17	momentum (12-1 month, honest execution)	primary	train	4,031	143	0.60%	0.55	-0.16%	rejected
H18	mean_reversion re-test	delivery	train	6,408	203	-0.44%	2.03	-0.10%	rejected
H19	oi_momentum	delivery	train	497	129	-1.10%	-2.77	-1.05%	rejected
H20	price_action LONG	delivery	train	3,389	210	-0.38%	-1.87	-0.19%	rejected
H21	momentum (12-1 month, honest execution)	delivery	train	1,367	49	2.02%	1.58	0.64%	accepted
H22	pairs_reversion	delivery	train	161	109	0.53%	0.85	1.06%	rejected
H23	momentum (12-1 month, honest execution)	delivery	val	233	9	6.55%	0.98	4.29%	accepted

Not shown: cdd796d6e171, an earlier pairs_reversion (primary/train) run superseded same-day by 0b11b017cef9 after a rollforward-at-entry fix (row H9 above) - its raw trades were overwritten on disk by the re-run and are not separately recoverable.

Monte Carlo block-bootstrap results

prob(mean>0) is the share of 10,000 resampled histories where the mean stays positive - read it as the bootstrap uncertainty around the real mean, not an independent probability the strategy is “truly” profitable.

#	Signal	Split/Window	prob(mean>0)	mean 95% CI	prob(compounded>0)
H1	mean_reversion re-test	primary/train	23.74%	-0.64% to 0.29%	10.76%
H2	delivery_breakout	delivery/train	4.40%	-1.36% to 0.07%	3.23%
H3	oi_momentum	primary/train	0.01%	-1.90% to -0.62%	0.00%
H4	participant_tilt	delivery/train	97.57%	0.00% to 2.14%	95.94%
H5	vol_squeeze_breakout	primary/train	0.00%	-1.50% to -0.70%	0.00%
H6	vol_squeeze_breakout (delay=2, honest execution)	primary/train	7.63%	-0.82% to 0.12%	3.12%
H8	price_action LONG	primary/train	0.00%	-1.63% to -0.72%	0.00%
H9	pairs_reversion	primary/train	81.23%	-0.17% to 0.57%	76.67%
H10	same_sector_pairing – random	primary/train	99.92%	0.11% to 0.45%	99.84%
H11	same_sector_pairing – liquidity	primary/train	99.99%	0.12% to 0.44%	99.95%
H12	same_sector_pairing – random	primary/val	73.81%	-0.15% to 0.28%	69.15%
H13	same_sector_pairing – liquidity	primary/val	73.44%	-0.17% to 0.30%	69.39%
H14	same_sector_pairing – random	delivery/train	63.79%	-0.17% to 0.25%	60.15%
H15	same_sector_pairing – liquidity	delivery/train	47.57%	-0.24% to 0.23%	43.50%
H16	vol_squeeze_breakout	delivery/train	21.23%	-0.77% to 0.34%	14.57%
H17	momentum (12-1 month, honest execution)	primary/train	70.70%	-1.07% to 1.95%	49.46%
H18	mean_reversion re-test	delivery/train	96.87%	-0.03% to 1.56%	94.45%
H19	oi_momentum	delivery/train	0.16%	-1.78% to -0.36%	0.02%
H20	price_action LONG	delivery/train	6.17%	-1.06% to 0.13%	4.12%
H21	momentum (12-1 month, honest execution)	delivery/train	94.26%	-0.37% to 3.83%	89.81%
H22	pairs_reversion	delivery/train	56.08%	-0.40% to 0.52%	50.97%
H23	momentum (12-1 month, honest execution)	delivery/val	81.53%	-4.46% to 17.72%	79.24%

Not covered: H7 (pairs_reversion pre-fix, same as above - no raw trades survive on disk).

Hypothesis diagnosis matrix

Every metric computed directly from the real saved trade files for each hypothesis - no estimation. CAGR / Sharpe / max drawdown come from the Rs 50,000, 5-slot portfolio simulation and are marked — where none exists (pairs trades have no portfolio-level simulation built in this project - noted explicitly in pairs_simulate.py's own docstring). “Short-only result” is — for every long-only single-leg signal by design (no honest short simulator was built for those). “Costs as % of gross P&L;” is shown against the ABSOLUTE gross figure so it stays interpretable when gross itself is negative; an asterisk flags hypotheses where gross P&L; is close enough to zero that the ratio is not meaningfully interpretable. Bull / bear / sideways use NIFTY50's own trailing 63-session return at each trade's entry date, with a +/-5% deadband for “sideways” – a descriptive convention for this report, not a re-fitted parameter.

Metric	H1	H2	H3	H4	H5	H6
Number of trades	20,499	2,194	2,448	2,916	4,792	4,802
Win rate	45.4%	47.0%	41.7%	48.3%	43.3%	46.5%
Avg winner	6.68%	4.88%	6.69%	6.56%	6.28%	6.13%
Avg loser	-7.33%	-4.19%	-5.86%	-7.12%	-5.48%	-5.19%
Profit factor	0.76	1.03	0.82	0.86	0.88	1.03
Gross expectancy	-0.650%	0.400%	-0.305%	-0.193%	-0.060%	0.400%
Net expectancy	-0.972%	0.077%	-0.627%	-0.515%	-0.382%	0.077%
Avg holding period (days)	8.6	8.8	8.5	8.8	8.4	8.9
Max drawdown	-75.3%	-25.9%	-73.5%	-38.0%	-70.2%	-78.2%
CAGR	-9.49%	0.38%	-9.06%	0.26%	-8.31%	-7.21%
Sharpe	-0.761	0.088	-0.739	0.072	-0.564	-0.276
Costs as % of |gross P&L;|	49.4%	80.8%	105.3%	166.5%	538.3%*	80.7%
Long-only result	-0.972%	0.077%	-0.627%	-0.515%	-0.382%	0.077%
Short-only result	—	—	—	—	—	—
Bull-market result	-0.969%	0.411%	-0.110%	0.072%	-0.398%	0.270%
Bear-market result	-0.505%	-0.527%	-2.655%	-0.949%	-1.563%	-0.672%
Sideways-market result	-1.453%	-0.374%	-0.728%	-0.674%	-0.087%	-0.133%

H1 = mean_reversion re-test (primary/train) · H2 = delivery_breakout (delivery/train) · H3 = oi_momentum (primary/train) · H4 = participant_tilt (delivery/train) · H5 = vol_squeeze_breakout (primary/train) · H6 = vol_squeeze_breakout (delay=2, honest execution) (primary/train)

Metric	H8	H9	H10	H11	H12	H13
Number of trades	8,807	216	3,409	3,705	2,068	2,301
Win rate	42.6%	61.1%	59.8%	59.7%	58.8%	56.9%
Avg winner	6.87%	1.47%	2.39%	2.18%	2.20%	2.17%
Avg loser	-6.14%	-1.86%	-2.87%	-2.62%	-3.01%	-2.72%
Profit factor	0.83	1.24	1.24	1.23	1.04	1.06
Gross expectancy	-0.269%	0.689%	0.787%	0.755%	0.565%	0.579%
Net expectancy	-0.591%	0.176%	0.275%	0.243%	0.052%	0.066%
Avg holding period (days)	8.4	9.5	9.7	9.8	10.1	10.1
Max drawdown	-73.1%	—	—	—	—	—
CAGR	-9.45%	—	—	—	—	—
Sharpe	-0.871	—	—	—	—	—
Costs as % of |gross P&L;|	119.6%	74.4%	65.1%	67.8%	90.8%	88.6%
Long-only result	-0.591%	0.218%	0.114%	-0.038%	0.182%	0.251%
Short-only result	—	0.134%	0.436%	0.524%	-0.078%	-0.119%
Bull-market result	-0.179%	0.434%	0.362%	0.328%	-0.179%	-0.318%
Bear-market result	-1.827%	-0.181%	0.541%	0.501%	0.689%	0.732%
Sideways-market result	-0.749%	0.076%	0.035%	0.022%	0.173%	0.407%

H8 = price_action LONG (primary/train) · H9 = pairs_reversion (primary/train) · H10 = same_sector_pairing – random (primary/train) · H11 = same_sector_pairing – liquidity (primary/train) · H12 = same_sector_pairing – random (primary/val) · H13 = same_sector_pairing – liquidity (primary/val)

Metric	H14	H15	H16	H17	H18	H19
Number of trades	1,739	2,021	1,725	4,031	6,408	497
Win rate	57.6%	55.8%	45.2%	53.5%	48.2%	37.8%
Avg winner	2.03%	2.11%	5.30%	11.22%	5.91%	5.72%
Avg loser	-2.93%	-2.80%	-4.70%	-11.64%	-6.35%	-5.25%
Profit factor	0.94	0.95	0.93	1.11	0.87	0.66
Gross expectancy	0.438%	0.451%	0.145%	0.925%	-0.119%	-0.776%

Metric	H14	H15	H16	H17	H18	H19
Net expectancy	-0.074%	-0.061%	-0.178%	0.601%	-0.441%	-1.097%
Avg holding period (days)	10.1	10.3	8.5	31.0	8.7	8.4
Max drawdown	—	—	-41.0%	-39.2%	-53.5%	-42.7%
CAGR	—	—	-2.00%	1.91%	-2.28%	-2.71%
Sharpe	—	—	-0.200	0.198	-0.149	-0.408
Costs as % of |gross P&L;|	116.9%	113.6%	223.1%*	35.0%	271.4%*	41.4%
Long-only result	-0.308%	-0.177%	-0.178%	0.601%	-0.441%	-1.097%
Short-only result	0.160%	0.054%	—	—	—	—
Bull-market result	-0.124%	-0.256%	0.152%	1.167%	0.548%	-0.691%
Bear-market result	0.245%	0.602%	0.131%	1.273%	-0.612%	-2.050%
Sideways-market result	-0.172%	-0.110%	-0.800%	-0.519%	-0.950%	-1.517%

H14 = same_sector_pairing – random (delivery/train) · H15 = same_sector_pairing – liquidity (delivery/train) · H16 = vol_squeeze_breakout (delivery/train) · H17 = momentum (12-1 month, honest execution) (primary/train) · H18 = mean_reversion re-test (delivery/train) · H19 = oi_momentum (delivery/train)

Metric	H20	H21	H22	H23
Number of trades	3,389	1,367	161	233
Win rate	44.9%	54.7%	55.9%	43.3%
Avg winner	5.62%	10.09%	2.38%	31.63%
Avg loser	-5.28%	-7.74%	-1.81%	-12.64%
Profit factor	0.87	1.58	1.66	1.92
Gross expectancy	-0.059%	2.342%	1.039%	6.885%
Net expectancy	-0.381%	2.016%	0.528%	6.552%
Avg holding period (days)	8.5	30.8	9.9	93.8
Max drawdown	-58.2%	-57.2%	—	-23.8%
CAGR	-3.54%	6.83%	—	0.77%
Sharpe	-0.290	0.446	—	0.146
Costs as % of |gross P&L;|	544.1%*	13.9%	49.2%	4.8%

Metric	H20	H21	H22	H23
Long-only result	-0.381%	2.016%	-0.264%	6.552%
Short-only result	—	—	1.320%	—
Bull-market result	0.341%	2.132%	0.617%	-1.242%
Bear-market result	-2.294%	4.714%	-0.306%	1.682%
Sideways-market result	-0.639%	0.405%	0.787%	9.078%

H20 = price_action LONG (delivery/train) · H21 = momentum (12-1 month, honest execution) (delivery/train) · H22 = pairs_reversion (delivery/train) · H23 = momentum (12-1 month, honest execution) (delivery/val)

Synthesis

• Every confirmation-style signal failed the same way. mean_reversion, delivery_breakout, oi_momentum, participant_tilt, vol_squeeze_breakout, and price_action all react to an already-visible dislocation, and all show the same mechanism: an inflated stop-hit rate vs. placebo, confirmed directly by a delay sweep on vol_squeeze_breakout (mean_net_pct improved monotonically with entry delay, turning positive at delay=2, though the portfolio-level drawdown got WORSE, not better - entry timing alone does not create a tradeable edge).

• “Smarter” filters did no better, sometimes worse. Correlation-screened pairs lost to random same-sector pairs; delivery/OI/flow overlays underperformed simpler variants. More granular conditioning data did not add information on this universe.

• Costs are usually the difference between “looks real” and “isn't”. Correlation-screened pairs' gross t-stat of 2.65 collapsed to 0.18 once honestly filled and costed - real round-trip costs (~0.3–0.6% single-leg, ~0.5% two-leg) exceed most gross edges found here.

• Large-n, high-t-stat single-window results are not enough on their own. same_sector_pairing's t=3.49–3.70 on 3,400+ trades (not a small-sample fluke) still collapsed on val and flipped sign entirely on the delivery split - concrete evidence for why the train/val/test discipline exists at all.

• Structural/portfolio failure is a separate axis from signal failure. mean_reversion's concentration risk (best per-trade expectancy, worst portfolio Sharpe, because correlated names crash together) and bear-gating's directionally-correct-but-insufficient regime story each fail for reasons unrelated to whether the entry itself carries edge.

• Plain NIFTY50 buy-and-hold beat every mechanical rule tested, on every window - consistent with obvious, bhavcopy-derivable signals (price, volume/delivery, OI, FII flow, volatility, sector correlation) being priced in or too thin to survive retail costs.

Open threads – not yet tried

Exit-geometry sweep on the other signals. A 125-cell hold/stop/risk-reward grid ruled out exit tuning as an explanation for mean_reversion's failure - never repeated on the other 5+ signals.

Entry-delay sweep on mean_reversion. The delay sweep that found the entry-timing mechanism only ever ran on vol_squeeze_breakout, despite mean_reversion having the most data of any signal in the project.

Expiry-matched holding period for pairs. Whether a holding period matched to the futures expiry cycle (enter right after a roll, exit before the next) would further reduce the 13–37% forced-rollover-exit rate is untested.

Volatility/credit stress-regime gate. Trailing-return regime (Phase 0) and FII-flow regime (participant_tilt) were each tried and rejected; a systemic-stress axis (India VIX term structure, credit spread proxy) never was. market_gate's own credit_spreads.py and vix_term_structure.py signals exist, are fully written, and are unused anywhere.

Fundamentals / valuation signals. Everything tested in this project so far is short-horizon and reactive to a visible dislocation. A valuation-based signal is a structurally different bet, but no fundamentals data pipeline exists in this project yet.

Calendar / seasonality effects. Never touched inside this project's own rigorous harness (market_gate has a separate, less rigorously-tested Seasonality page).

Known limitations of this report

• No portfolio-level simulation exists for pairs trades (same_sector_pairing, pairs_reversion) – CAGR/Sharpe/max-drawdown cells are blank by design, not missing data; margin/leverage economics for a multi-pair book were never modelled.

• price_action's SHORT side was only ever screened (no cost model, approximate same-day-close fills, no honest simulator) – excluded from this matrix entirely rather than mixed in under a different methodology than every other cell.

• The bull/bear/sideways split in the matrix is a description for this report, built after the fact – it was never used to gate or select a real hypothesis, and should not be read as a fourth confirmed regime-conditional edge.

• “Accepted” in this report always means “cleared this project's own mechanical logging bar”, never “confirmed” – every acceptance here has either failed its own val test or remains thin enough (single-digit bucket counts, t<1) that it should not be traded on this evidence alone.